21.6. $\Delta x \Delta p_x \geq \frac{\hbar}{2}$

$\Delta p_x = m\Delta v$

$= 9.1\times10^{-31}\,kg \times 1\times10^{7}\,m/s \times 1\%$

$= 9.1\times10^{-26}\,kg\cdot m/s$

$\Delta x \geq \frac{1.05\times10^{-34}}{2\times9.1\times10^{-26}\,kg\cdot m/s} = 5.8\times10^{-10}\,m$

21.10. $\psi(x) = \sqrt{\frac{2}{a}}\sin\frac{3\pi x}{a}$

$|\psi(x)|^2 = \frac{2}{a}\sin^2\frac{3\pi x}{a}$

$\frac{d|\psi(x)|^2}{dx} = \frac{2}{a}\,2\sin\frac{3\pi x}{a}\cdot\cos\frac{3\pi x}{a}\times\frac{3\pi}{a}$

$= \cancel{\frac{12\pi}{a^2}}\cdot\frac{6\pi}{a^2}\cdot\sin\frac{6\pi}{a}x = 0$

$\frac{6\pi}{a}x = n\pi$，则 $x = \frac{na}{6}$，$n\in Z$.

又 $0<x<a$，故 $x=\frac{a}{6}, \frac{a}{3}, \cancel{\frac{a}{2}}, \frac{a}{2}, \frac{2}{3}a, \frac{5}{6}a$

$x=\frac{a}{3}, \frac{2}{3}a$ 时，$|\psi(x)|^2=0$，为极小值，舍去.

故出现概率密度最大的各位置为 $x=\frac{a}{6}, \frac{a}{2}, \frac{5}{6}a$.

21.11 $\psi = cx(l-x)$.

$\int_0^l |\psi(x)|^2 dx = \int_0^l (cxl - cx^2)^2 dx = \int_0^l (c^2x^2l^2 + c^2x^4 - 2c^2x^3l)dx$

$= \left(\frac{c^2l^2}{3}x^3 + \frac{c^2}{5}x^5 - \frac{2c^2l}{4}x^4\right)\Big|_0^l = \frac{c^2l^5}{3} + \frac{c^2l^5}{5} - \frac{c^2l^5}{2} = 1$，得 $c^2 = \frac{30}{l^5}$

则 $c=\sqrt{\frac{30}{L^5}}$.

则 $\int_0^{\frac{L}{3}}|\psi(x)|^2dx=\frac{30}{L^5}\left(\frac{L^2}{3}x^3+\frac{x^5}{5}-\frac{Lx^4}{2}\right)\Big|_0^{\frac{L}{3}}$

$$=\frac{30}{L^5}\cdot\left(\frac{L^5}{81}+\frac{L^5}{243\times5}-\frac{L^5}{2\times81}\right)$$

$$=\frac{17}{81}$$

则 $P=\frac{17}{81}$

21.13. $\Delta x\cdot\Delta p_x\geq\frac{\hbar}{2}$

$\Delta p_x=m\Delta v\geq\frac{\hbar}{2\times L}$

则 $\Delta v\geq\frac{\hbar}{2mL}$

粒子具有最小能量: $E_{min}=\frac{1}{2}\times m\times\left(\frac{\hbar}{2mL}\right)^2=\frac{\hbar^2}{8mL^2}=\frac{h^2}{32\pi^2mL^2}$

质子中子最小能量: $E_{pmin}=\frac{h^2}{32\pi^2\times1.67\times10^{-27}kg\times(10^{-14}m)^2}$

$$=8.3\times10^{-15}J.$$

21.16. $a=0.6nm$. $E_n=n^2\frac{\pi^2\hbar^2}{2ma^2}$

$n=1,2,3$. $E_1=\frac{\pi^2\hbar^2}{2\times(9.1\times10^{-31}kg)\times(0.6nm)^2}=1.05eV$

$E_2=4E_1=4.2eV$.

$E_3=9E_1=9.45eV$.

21.17. (1). $\psi(x)=\begin{cases} Axe^{-\lambda x} & (x\geq 0) \\ 0, & (x<0) \end{cases}$

$$\int_0^\infty |\psi(x)|^2 dx = \int_0^\infty A^2x^2e^{-2\lambda x}\,dx.$$

$$= A^2\cdot\left(\frac{x^2}{-2\lambda}\cdot e^{-2\lambda x}\Big|_0^\infty + \int_0^\infty \frac{1}{\lambda}xe^{-2\lambda x}dx\right)$$

$$= A^2\int_0^\infty \frac{1}{4\lambda^3}\cdot(-2\lambda x)e^{-2\lambda x}\,d(-2\lambda x)$$

$$= \frac{A^2}{4\lambda^3}\cdot(-2\lambda x-1)\cdot e^{-2\lambda x}\Big|_0^\infty = \frac{A^2}{4\lambda^3} = 1.$$

则 $A = 2\lambda^{\frac{3}{2}}$.

则 $\psi(x) = \begin{cases} 2\lambda^{\frac{3}{2}}xe^{-\lambda x} & (x\geq 0) \\ 0, & (x<0) \end{cases}$

(2). $|\psi(x)|^2 = \begin{cases} 4\lambda^3 x\cdot e^{-2\lambda x} & (x\geq 0) \\ 0, & (x\leq 0) \end{cases}$

21.18. $e^{-2ka} = \frac{1}{100}$, $2ka = 4.6$.

$a = \frac{2.3}{k}$, $k = \sqrt{\frac{8\pi^2\times 9.1\times 10^{-31}kg\times 1eV}{h^2}}$

$= 5.12\times 10^9$.

$a = 4.5\times 10^{-10}m = 0.45nm$